

DET-013 | SMB Lateral Movement to Multiple Hosts

ID: DET-013

MITRE ID: T1021.002

Data Source: DeviceNetworkEvents

Severity: High

Tactic:Lateral Movement

Category:LATERAL MOVEMENT

ADX MAPPING LOGIC:

Mapping: State = 'source device', Source = 'destination IP', DamageProperty = 'bytes transferred proxy'. We use make-series and series_decompose_anomalies() to detect anomalous spikes — advanced operators.

OBJECTIVE: Practice make-series and series_decompose_anomalies() — advanced time-series anomaly detection operators. Real Sentinel: DeviceNetworkEvents port 445 connections to many IPs from one host.

KQL QUERY (RUNNABLE ON ADX FREE TIER)

```
StormEvents
| where StartTime > ago(365d)
| make-series
ConnectionCount = count(),
TotalDamage = sum(DamageProperty)
on StartTime
from ago(365d) to now() step 7d
by State // State = source device
| extend Anomalies = series_decompose_anomalies(ConnectionCount)
| mv-expand
StartTime,ConnectionCount,
TotalDamage,
Anomalies
| extend
IsAnomaly = toint(Anomalies) > 1, // score >1 = anomalous spike
AnomalyScore = toint(Anomalies)
| where IsAnomaly == true
| project StartTime = todatetime(StartTime),DeviceName = State,
ConnectionCount = toint(ConnectionCount),TotalDamage = tolong(TotalDamage),
AnomalyScore,IsAnomaly
| order by AnomalyScore desc
| take 20
```

INVESTIGATION STEPS

1. Identify if the source device is a file server (expected high SMB).
2. Check if SMB connections go to known internal file shares.
3. Review process activity on the source device at the same time.
4. Check for EternalBlue or other SMB exploit patterns

RESPONSE ACTIONS

- Isolate the source device from the network.
- Block SMB (port 445) outbound at the firewall for that host.
- Scan all target hosts for signs of compromise.
- Check for ransomware indicators (file encryption activity).

Output from Above KQL Query:

```
1 StormEvents
2 | where StartTime > ago(365d)
3 | make-series ConnectionCount = count(), TotalDamage = sum(DamageProperty)
4 on StartTime from ago(365d) to now() step 7d
5 by State // State = source device
6 | extend Anomalies = series_decompose_anomalies(ConnectionCount)
7 | mv-expand
8 StartTime,
9 ConnectionCount,
10 TotalDamage,
11 Anomalies
12 | extend
13 IsAnomaly = toint(Anomalies) > 1, // score >1 = anomalous spike
14 AnomalyScore = toint(Anomalies)
15 | where IsAnomaly == true
16 | project StartTime = todatetime(StartTime), DeviceName = State,
17 ConnectionCount = toint(ConnectionCount), TotalDamage = tolong(TotalDamage),
18 AnomalyScore, IsAnomaly
19 | order by AnomalyScore desc | take 20
```

Table 1 + Add visual Stats

StartTime	DeviceName	ConnectionCount	TotalDamage	AnomalyScore	IsAnomaly
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No Rows To Show

KQL Detection Query in Sentinel:

DeviceNetworkEvents

| where TimeGenerated > ago(1h)

| where RemotePort == 445

| where ActionType == "ConnectionSuccess"

| summarize

TargetHosts = dcount(RemoteIP),

HostList = make_set(RemoteIP),

TotalConns = count(),

FirstSeen = min(TimeGenerated),

LastSeen = max(TimeGenerated)

by DeviceName, LocalIP

| where TargetHosts > 10

| extend ConnsPerMin = TotalConns / (datetime_diff('minute', LastSeen, FirstSeen) + 1)

| project DeviceName, LocalIP, TargetHosts, TotalConns, ConnsPerMin, HostList,

FirstSeen, LastSeen

| order by TargetHosts desc